AAE/IE 561: *Introduction to convex optimization*

Term: 2025 Fall

1. Course Information

- **Time and location:** Monday 4:30 PM – 7:20 PM at Grissom Hall 134, in person.
- **Credits:** 3.

2. Course Description

Optimization is a broad mathematical discipline that has achieved widespread use in many applied sciences, including machine learning, signal processing, statistics, and various applications in engineering. Beginning with classical roots in the calculus of variations, the subject has flourished: from the celebrated simplex method in linear programming to the mature theories of duality and algorithmic complexity in convex optimization, it has now enjoyed over 60 years of theoretical and computational advances. Key to these advances were the development of convex analysis and the recognition of its role in first-order optimality conditions. Based on duality, nonsmooth calculus, and the techniques of numerical linear algebra, the subject now enjoys an extensive algorithmic toolbox, containing practical and provably efficient numerical algorithms.

This course provides an introduction to the theory of convex optimization, aiming to develop a deep understanding of convexity and the principles underlying algorithms for convex optimization problems. Emphasis will be placed on recognizing, formulating, analyzing, and solving convex optimization problems.

While applications will be included in the class, the course is not application driven but heavily **math oriented.** All results will be provided along with rigorous proofs, and **you are expected to write lots of math proofs.**

3. Course references

- Convex optimization by Boyd and Vandenberghe
- Numerical optimization by Nocedal and Wright
- Lectures on convex optimization by Nesterov

4. Topics covered in this course and learning outcomes

This course would contain two parts: basic convex analysis and convex optimization algorithms. In the first half of this course, we introduce convexity and theoretically show various consequence of convexity, which are helpful for algorithm development. In particular, the following will be covered:

1. Definition of convex sets/functions, and how to recognize them.
2. Normal cone, first-order optimality conditions, and Separating hyperplane.
3. Lagrangian duality, KKT conditions.

4. Nonsmooth calculus: subdifferential and their existence.

In the second half of the course, we introduce algorithms that solves convex optimization problems and prove their convergence guarantees. Algorithms we intend to cover:

1. First-order methods: gradient descent, subgradient method, proximal point method, stochastic gradient descent, (optional) accelerated gradient descent.
2. Beyond first-order methods: Newton's method, (optional) BFGS
3. Algorithms for problems with complex constraints: Augmented Lagrangian, ADMM.

You will have solid foundation in convex optimization after taking this course.

5. Assignments and Exam

Homework will be assigned roughly bi-weekly. The homework problems are designed to reinforce the material introduced to you in lecture, on your own time. The typical homework consists of proving some given property (e.g., convexity of a set, convexity of a function) or statement (e.g., the closure of a convex set is still convex), or provide counter examples (e.g., provide an example of a continuous function which does not have a minimizer). In the second part of the course, there might be some coding assignments. Solutions to the homework will be posted on Brightspace. **You have 4 late days in total, and can use at most 2 late days for each individual homework. The homework with the lowest score percentage will be dropped when calculating final grades.**

You are encouraged to work with other students. Discussing the course material and homework assignments is a great way to learn, **but you have to write your homework yourself** and **acknowledge all the help you received (including specifying which questions you got help)** clearly at the beginning of each assignment. Failure to do so is a violation of academic integrity.

There will be an 90-min in class exam on October 27th. The exam will be open book and open notes (no electronic device allowed).

6. Grading Scheme

We will use an ingenious grading scheme (originating from Ben Grimmer). Course grades are based on four components: **Homework**, **Exam**, and **Participation**. For each student, I will select the rubric (weights) that maximizes their course score, subject to reasonable constraints.

Component scores. Let

$$C_H = \text{Homework score}, \quad C_E = \text{Exam score}, \quad C_P = \text{Participation score},$$

each on a 0–100 scale.

Weights (rubric). Let

$$H = \text{Homework weight}, \quad E = \text{Exam weight}, \quad P = 100 - H - E \quad (\%; \text{ weights sum to } 100).$$

Per-student optimization. For each student, the course score is obtained by solving

$$\begin{aligned}
\max_{H,E,F} \quad & \frac{1}{100}(C_H H + C_E E + C_P\big(100 - H - E\big)) \\
\text{s.t.} \quad & (H, E) \in \mathbb{R}^2, \\
& H + E \leq 100 && \text{(percentages are at most 100)}, \\
& 30 \leq H,\ E && \text{(Homework and Exam matter)}, \\
& E \leq H && \text{(Doing well on homeworks is more important than doing well in one exam)}, \\
& 90 \leq H + E && \text{(Homework and exam are the majority of the score)}.
\end{aligned}$$

The optimal weights $(H^\star, E^\star, P^\star)$ determined by this problem are then used to compute the student's final course score.

In Table 1, you will see our grading scale. **Note that it should be understood as a guarantee, i.e., once you achieve the grade range, you are guaranteed to get the corresponding letter grade or a even better grade.** I have the right to lower the threshold for specific letter grades.

Grade	Score Range
A+	≥ 95
A	[90, 95)
A−	[86.6, 90)
B+	[83.3, 86.6)
B	[80, 83.3)
B−	[76.6, 80)
C+	[73.3, 76.6)
C	[70, 73.3)
C−	[66.6, 70)
D+	[63.3, 66.6)
D	[60, 63.3)
D−	[56.6, 60)
F	< 56.6

Table 1: Grading scale

There is also a grade for "ethics." This is **Pass-Fail**. Not cheating and not being involved in any other unethical behavior according to the student honor code will be considered Pass. (Otherwise, it is a failure). Failing ethics will result in you getting an F for the course.